A Finite Polylogarithm Alphabet for Rational Hankel Sums

Confluent pole coefficients, Gamma-product normalizations, endpoint cancellation, and Stieltjes spectral identities

Research prepared for Vladimir Reshetnikov
Mathematical derivations and computational checks with ChatGPT

11 October 2026

Abstract

We give an explicit finite multiple-polylogarithm reduction for every cutoff Hankel determinant of a rational weight whose poles are at nonpositive integers. The number of particles may be arbitrary. The controlling invariant is not determinant size or ordinary harmonic rank: a pole of multiplicity s supplies the decreasing positive sequence $s, s-2, s-4, \ldots$. The sum of the r largest available entries bounds the polylogarithmic weight for an r-particle determinant. A confluent Vandermonde factorization proves this statement; a Hermite-reduction argument turns it into a terminating identity compiler. For a single pole the formal bound is attained at every rank, and a specified extremal coefficient is an explicit product of Gamma factors. For several simple poles we derive a fixed-size Hahn determinant for the highest harmonic coefficient, a complete finite formula for each endpoint principal part, and an exact spectral description of one-residue exceptional parameters. A separate spectral determinant identity yields ordinary all-order relations involving generalized Stieltjes constants and derivatives of depth-two and higher zeta functions. All infinite identities are proved analytically; exact finite tests and numerical diagnostics are supplied separately. Historical priority beyond the inspected sources is not claimed.

Main distinction. "Weight" below is a grading of an explicit expression, not a claim of transcendence, algebraic independence, or minimal multiple-polylogarithm depth. The unrestricted harmonic-tail moment-prefix conjecture and the manuscript's Gaussian S_6 and revised S_8 identities are not resolved here.

Contents

1 Research scope and comparison with the repository **3**
1.1 A fixed comparison point and a moving repository 3

2 Conventions and the exact object to be reduced **3**

3 The confluent pole-multiplicity theorem **4**
3.1 A moment factorization by local derivative columns 4
3.2 The invariant and its closed combinatorial form 5

4 A sharp single-pole coefficient with Gamma normalization **6**

5 From the pole grading to a finite polylogarithm identity **7**
5.1 Cost-preserving cutoff partial fractions 7
5.2 Finite harmonic products and their generating functions 8
5.3 A closed rational integration step 8

6 Several simple poles: fixed-size highest coefficients **10**
6.1 Exceptional residues as an exact matrix spectrum 12

7 Exact endpoint cancellation equations **12**
7.1 A complete three-pole example 13

8 A small exact identity with a rank-independent extension **14**
8.1 The triple pole and its weight-four primitive 14
8.2 A shifted double pole with an ordinary zeta endpoint 16

9 A spectral master identity at every rank **16**

10 Generalized Stieltjes constants and exact spectral jets **18**
10.1 A complete rank-two Stieltjes family 18
10.2 The all-rank version at the Stieltjes boundary 19
10.3 Anchored spatial primitives and the Gamma endpoint 20

11 Verification, claim status, and source corrections **20**
11.1 Exact checks and numerical diagnostics have different roles 20
11.2 What the package implements 21
11.3 Corrections and integration decisions 21

12 Further research questions **22**

Conclusion **22**

1 Research scope and comparison with the repository

The natural starting point for this continuation is the several-pole question in the incoming gap-six report, rather than an older Gamma normal-form question that has already been answered. In the inspected *Gap Six Harmonic Determinants* package, Section 3 gives an arbitrary-particle formula for two adjacent simple poles using only a dilogarithm and powers of $-\log(1-z)$. Section 5, Questions Q7–Q9, asks for several-pole extensions, a classification of when classical polylogarithms suffice, and exact Euler primitives [2]. The same package already proves the complex rational-weight *ordinary harmonic degree–rank* theorem. We do not claim that result again.

The new contribution developed here is a different grading and its exact analytic realization. Repeated poles make ordinary harmonic degree insufficient: $H_M^{(1)}H_M^{(3)} - (H_M^{(2)})^2$ has harmonic degree two but weight four. Our confluent minor calculation gives a uniform finite alphabet, explicitly tracks the cutoff poles needed by the compiler, and supplies a nonzero extremal coefficient for the single-pole family. The several-simple-pole result gives more than a bare appeal to a Christoffel transformation: it specifies the compressed measure, the normalization, the principal-part cancellation equations, and a positive-matrix description of exceptional residue parameters.

Several ingredients are classical. Finite Cauchy–Binet, confluent Vandermonde matrices, finite harmonic quasi-shuffles [6], Christoffel determinant transformations [7], uniform Hahn polynomials [10], and integration of hyperlogarithms against rational functions [8] are used with their original status intact. Determinantal stability is also an established method [9]. The contribution is their particular combination with the pole-multiplicity invariant, its sharp coefficient, and the resulting identity families. Neither the software architecture for hyperlogarithmic integration nor the classical weight-five Euler sums used in an example are claimed as new.

1.1 A fixed comparison point and a moving repository

During retrieval, `main` changed while older incoming archives were being placed in report directories. The final observation used here is the exact commit

`505c62fddea6b4a9eb61fee0c76e6c021e9c7128`.

The gap-six archive at that commit has Git blob `d7695b94c3117d7bdcb5e75fcf930c410e9c31b6`. Its sections on complex rank, adjacent-pole generators, and further questions were read directly from the archive. The universal Gamma normal-form and Gamma-normalization archives were also checked to avoid reopening their completed questions [3, 4]. The source audit accompanying this article identifies the scope of each inspection. It is not an assertion that every page in the repository or every incoming archive was reviewed. No repository file was modified.

2 Conventions and the exact object to be reduced

Let $r \geq 1$ and $M \geq 0$ be integers. We use the nodes $1, \dots, M$ and write

$$R(x) = P(x) + \sum_{a \in A} \sum_{\ell=1}^{s_a} \frac{c_{a,\ell}}{(x+a)^\ell}, \qquad A \subset \mathbb{Z}_{\geq 0} \text{ finite.} \tag{2.1}$$

Here P is a polynomial, $c_{a,s_a} \neq 0$, and coefficients may be complex. All denominators are therefore nonzero on the summation nodes. The field K is generated over $\mathbb{Q}$ by the coefficients of R. Set

$$\mu_j(M) = \sum_{n=1}^{M} n^j R(n),$$
$$D_{r,M}(R) = \det(\mu_{i+j}(M))_{0\le i,j<r}, \qquad G_r(z;R) = \sum_{M\ge 1} D_{r,M}(R) z^M. \tag{2.2}$$

The empty-node matrix gives $D_{r,0} = 0$. In fact $D_{r,M} = 0$ when $M < r$. Finite Cauchy–Binet gives the exact identity

$$D_{r,M}(R) = \sum_{1\le n_1<\cdots<n_r\le M} \prod_{i<j} (n_j - n_i)^2 \prod_{j=1}^{r} R(n_j). \tag{2.3}$$

The source's zero-based nodes $0, \dots, N$ translate to these conventions by $M = N+1$, $n_{\text{source}} = n-1$, and $a_{\text{source}} = a+1$. A monic affine change of polynomial basis has determinant one, so it preserves the Hankel determinant. There is no factor $r!$ in this ordered sum. Expansion of the two Vandermonde determinants proves (2.3) directly. The terms grow at most polynomially in M, so (2.2) converges absolutely for $|z| < 1$, locally uniformly there.

The generalized harmonic sums are

$$H_M^{(p)} = \sum_{n=1}^{M} n^{-p}, \qquad h_{a,p}(M) = \sum_{n=1}^{M} (n+a)^{-p}. \tag{2.4}$$

A polynomial expression in formal symbols $h_{a,p}$ has *weighted harmonic degree* obtained by assigning $\mathrm{wt}(h_{a,p}) = p$ and weight zero to coefficients depending polynomially on M. This grading is first imposed before identifying the shifted harmonic symbols. For integral $a \ge 0$, the exact shift rule is

$$h_{a,p}(M) = H_M^{(p)} + \sum_{b=1}^{a} \frac{1}{(M+b)^p} - H_a^{(p)}. \tag{2.5}$$

Thus an inverse cutoff factor of order p inherits a cost of p. The constants $H_a^{(p)}$ are rational and have cost zero.

We use the decreasing-index convention

$$\mathrm{Li}_{p_1,\dots,p_k}(z) = \sum_{n_1>\cdots>n_k\ge 1} \frac{z^{n_1}}{n_1^{p_1}\cdots n_k^{p_k}}, \qquad p_i \in \mathbb{Z}_{>0}. \tag{2.6}$$

Its weight is $p_1 + \cdots + p_k$. Denote by $\mathscr{L}_W$ the span over $K(z)$ of 1 and all these functions of weight at most W. Every rational prefactor constructed below has poles only at 0 and 1. No assertion that the displayed generators are a minimal basis is needed.

3 The confluent pole-multiplicity theorem

3.1 A moment factorization by local derivative columns

For a fixed pole $-a$ of order $s = s_a$, define the $r \times s$ matrix

$$V_a(i,p) = \begin{cases} \binom{i}{p}(-a)^{i-p}, & p \le i, \\ 0, & p > i, \end{cases} \qquad 0 \le i < r, \quad 0 \le p < s. \tag{3.1}$$

The value 0^0 occurring here is one. Define a symmetric $s \times s$ harmonic matrix by

$$C_a(p,q) = \sum_{\ell=p+q+1}^{s} c_{a,\ell} h_{a,\ell-p-q}. \tag{3.2}$$

An empty sum is zero. In particular $\operatorname{wt} C_a(p,q) \le s-p-q$ for a nonzero entry.

Lemma 3.1 (Confluent harmonic factorization)**.** *There is a matrix $B_r(M)$ with entries in $K[M]$ such that the formal moment matrix is*

$$\boldsymbol{\mu}_r(M) = B_r(M) + \sum_{a\in A} V_a C_a V_a^{\mathsf{T}}. \tag{3.3}$$

The matrix B_r is the finite sum of the polynomial parts of $x^{i+j}R(x)$. The transpose is ordinary transpose, not conjugate transpose.

Proof. Expand x^i and x^j at $x=-a$:

$$x^i = \sum_p \binom{i}{p} (-a)^{i-p}(x+a)^p.$$

After multiplying and dividing by $(x+a)^\ell$, terms with $p+q<\ell$ sum to $h_{a,\ell-p-q}$ and are exactly (3.2)–(3.3). The other terms are nonnegative powers of $x+a$ and have polynomial Faulhaber sums in M. Equivalently, the partial-fraction coefficient uses $\binom{i+j}{t} = \sum_{p+q=t} \binom{i}{p}\binom{j}{q}$. This proves every matrix entry, including the polynomial part of R. □

3.2 The invariant and its closed combinatorial form

For $r \ge 0$, define

$$W_r(\mathbf{s}) = \max_{\substack{0\le k_a\le s_a \\ \sum_a k_a \le r}} \sum_{a\in A} k_a(s_a - k_a + 1). \tag{3.4}$$

For no poles, $W_r = 0$.

Theorem 3.2 (Weighted harmonic degree)**.** *Every monomial in the formal determinant obtained from (3.3) has weighted harmonic degree at most $W_r(\mathbf{s})$. Equivalently, form the multiset*

$$\bigsqcup_{a\in A} \{s_a, s_a-2, s_a-4, \ldots\} \cap \mathbb{Z}_{>0}, \tag{3.5}$$

retaining multiplicities. Then W_r is the sum of the largest r entries, or all entries if there are fewer than r. In particular

$$W_r \le W_\infty := \sum_{a\in A} \left\lfloor \frac{(s_a+1)^2}{4} \right\rfloor. \tag{3.6}$$

These are bounds on an exact expression grading, not inequalities between values of special functions.

Proof. Concatenate the V_a and make the C_a into a block diagonal matrix C. In the expansion of $\det(B + VCV^{\mathsf{T}})$, choose k columns from the second summand and the complementary columns from B. Apply Cauchy–Binet to the resulting $k\times k$ minors of VCV^{T}. All dependence on harmonic symbols lies in minors $\det C[I,J]$ with $|I| = |J| = k \le r$.

A nonzero minor of this block diagonal matrix selects the same number k_a of row and column indices from every pole block. Let I_a, J_a be those local indices, each a subset of $\{0, \dots, s_a - 1\}$. Every term of its determinant has weight at most

$$k_a s_a - \sum_{p \in I_a} p - \sum_{q \in J_a} q \le k_a s_a - k_a(k_a - 1) = k_a(s_a - k_a + 1).$$

The second inequality uses the smallest possible sum of k_a distinct nonnegative integers, $k_a(k_a-1)/2$, separately for rows and columns. The V and B minors have weight zero. Adding the block contributions and maximizing proves (3.4).

The increase in $k(s-k+1)$ when k is replaced by $k+1$ is $s-2k$. These marginal increases decrease within each pole block. Maximizing with at most r choices therefore selects the largest positive marginals globally; their decreasing order automatically respects every block's prefix constraint. Zero marginals may be omitted and negative ones never help. Summing all positive marginals yields $\lfloor (s+1)^2/4 \rfloor$ for each block. This proves (3.5) and (3.6). □

For simple poles this specializes to $W_r = \min(r, |A|)$. For one pole of orders $1, 2, 3, 4, 5, 6, 7$, the stabilized weights are respectively $1, 2, 4, 6, 9, 12, 16$. The jump from order two to order three is real: pole multiplicity itself is not the correct weight budget.

Remark 3.3 (What cancellation can and cannot imply). The upper bound permits special residues to cancel its highest components. It does not assert weighted-degree equality for every multiple-pole complex weight. It also does not prove that a function of weight W_r cannot be written using functions of smaller weight. A precise formal sharpness theorem for a single pole follows next.

4 A sharp single-pole coefficient with Gamma normalization

Put $R(x) = x^{-s}$ with $s \ge 1$ integral. Polynomial division is trivial:

$$\mu_j(M) = \begin{cases} H_M^{(s-j)}, & j < s, \\ S_{j-s}(M), & j \ge s, \end{cases} \qquad S_t(M) = \sum_{n=1}^{M} n^t. \tag{4.1}$$

Let $\mathcal{P}_{r,s}(M; H_1, \dots, H_s)$ be this formal determinant. Write

$$\mathcal{U}_q(M) = \prod_{j=0}^{q-1} \frac{\Gamma(M+j+1)}{(2j+1)\Gamma(M-j)\binom{2j}{j}^2}, \qquad \mathcal{U}_0(M) = 1. \tag{4.2}$$

For $M \ge q$ all factors are finite and positive. They also have polynomial continuations because each Gamma quotient is $(M-j)_{2j+1}$.

Theorem 4.1 (An explicit nonzero extremal coefficient)**.** *Let*

$$k = \min(r, \lceil s/2 \rceil), \qquad q = r - k, \qquad W = k(s-k+1) = W_r(s).$$

The coefficient of H_{s-k+1}^k in $\mathcal{P}_{r,s}$ is

$$\boxed{[H_{s-k+1}^k]\mathcal{P}_{r,s} = (-1)^{k(k-1)/2}\mathcal{U}_q(M) \begin{cases} 1, & q = 0 \text{ or } s \text{ even}, \\ \dfrac{(M+1)_q}{\binom{2q}{q}}, & q > 0 \text{ and } s \text{ odd}. \end{cases}} \tag{4.3}$$

Consequently the formal weighted harmonic degree is exactly $W_r(s)$ as a polynomial in M, and also at every integer $M \ge r$.

Proof. A term contributing the displayed monomial must select k harmonic entries. Its harmonic weight is $k(s-k+1)$. Equality in the minor bound in Theorem 3.2 forces the chosen row and column sets to be $\{0,\dots,k-1\}$. Within that harmonic minor, the monomial with all indices $s-k+1$ requires $i+\sigma(i)=k-1$ at every row. The unique permutation is reversal and has sign $(-1)^{k(k-1)/2}$.

The complementary minor has indices $i,j\ge k$. If it is nonempty, then $k=\lceil s/2\rceil$ and $2k-s=\varepsilon\in\{0,1\}$. It is the polynomial moment determinant

$$\det\left(\sum_{n=1}^{M} n^{i+j+\varepsilon}\right)_{0\le i,j<q}. \tag{4.4}$$

For $\varepsilon=0$, its value is the product of the monic uniform Hahn norms, namely $\mathcal{U}_q(M)$. For $\varepsilon=1$, the one-factor Christoffel formula gives $(-1)^q\mathcal{U}_q(M)p_q(0;M)$, where p_q is monic and orthogonal on $1,\dots,M$ with unit weights. Its endpoint value is

$$p_q(0;M)=(-1)^q\frac{(M+1)_q}{\binom{2q}{q}}.$$

The finite Hahn formula and normalization are recalled in Section 6; they are classical. This proves (4.3). Positivity of (4.4) when $M\ge q$ proves nonvanishing in the stated range. The upper bound already proved then gives exact formal degree. When s is even, an additional zero-marginal choice can have the same weight and a different harmonic degree; it cannot contribute the monomial of degree k selected here. □

For example, the coefficients of H_2^2 for the triple pole are

$$\begin{aligned}
[H_2^2]\mathcal{P}_{2,3}&=-1,\\
[H_2^2]\mathcal{P}_{3,3}&=-\frac{M(M+1)}{2},\\
[H_2^2]\mathcal{P}_{4,3}&=-\frac{M^2(M-1)(M+1)^2(M+2)}{72}.
\end{aligned} \tag{4.5}$$

All ranks nevertheless have the same maximal weight four. This is a coefficient identity with an explicit Gamma-product source, not an asymptotic observation.

5 From the pole grading to a finite polylogarithm identity

5.1 Cost-preserving cutoff partial fractions

Lemma 5.1 (Normal form after merging integer shifts)**.** *After applying (2.5), $D_{r,M}(R)$ is a finite sum of terms of the following two types:*

$$p(M)\prod_{j=1}^{v}H_M^{(b_j)},\qquad \frac{c}{(M+h)^e}\prod_{j=1}^{v}H_M^{(b_j)}, \tag{5.1}$$

where $p\in K[M]$, $c\in K$, $h\in\{1,\dots,\max A\}$, $e\ge 1$, and

$$\sum_j b_j\le W_r\quad\text{in the first type},\qquad e+\sum_j b_j\le W_r\quad\text{in the second type}. \tag{5.2}$$

The empty harmonic product is one. If A is empty or $A=\{0\}$, there are no cutoff-denominator terms.

Proof. Before merging shifts, Theorem 3.2 supplies a polynomial in the $h_{a,p}$ with polynomial coefficients in M. Expanding each factor by (2.5), choose either a harmonic factor of weight p, an endpoint denominator of order p, or a rational constant. The total of retained harmonic weight and all denominator orders cannot increase. A rational coefficient with total denominator order E has only the listed poles, each of order at most E. Partial fractions and polynomial division give (5.1); every resulting individual pole order is at most E. This proves (5.2). No cancellation assumption is used. □

5.2 Finite harmonic products and their generating functions

For a word $\mathbf{b} = (b_1, \dots, b_v)$ of positive integers define

$$H_M(\mathbf{b}) = \sum_{M \geq n_1 > \cdots > n_v \geq 1} n_1^{-b_1} \cdots n_v^{-b_v}.$$

The empty word has value one. Partitioning a product of finite sums by which indices coincide gives

$$H_M(a)H_M(b) = H_M(a,b) + H_M(b,a) + H_M(a+b), \tag{5.3}$$

and the recursive rule

$$(a,u) * (b,v) = (a, u * (b,v)) + (b, (a,u) * v) + (a+b, u * v). \tag{5.4}$$

The empty word is the identity and identical output words retain their integer multiplicities. This is the positive finite-sum quasi-shuffle, not the version with a negative merger sign used for some star-sum conventions [6]. The rule preserves total weight. Furthermore, for every nonempty word,

$$\sum_{M \geq 1} H_M(\mathbf{b}) z^M = \frac{\mathrm{Li}_{\mathbf{b}}(z)}{1-z}. \tag{5.5}$$

Indeed the largest summation index n_1 is contained in every cutoff $M \geq n_1$; summing its geometric tail proves the formula. For the empty word the corresponding generator is $z/(1-z)$, not $1/(1-z)$. Absolute convergence for $|z| < 1$ justifies these rearrangements.

5.3 A closed rational integration step

Use hyperlogarithm words on the two letters $0, 1$, with kernels $f_0(z) = 1/z$ and $f_1(z) = 1/(1-z)$. Let $I_\emptyset = 1$ and, for a nonempty word ending in 1, set

$$I_{a\omega}(z) = \int_0^z f_a(t) I_\omega(t)\, \mathrm{d}t.$$

These anchored integrals are equivalent to (2.6):

$$\mathrm{Li}_{b_1,\dots,b_v}(z) = I_{0^{b_1-1}1\cdots 0^{b_v-1}1}(z). \tag{5.6}$$

Their weight is word length. Intermediate rational integrations can produce $\log z$; the following statement includes the necessary cancellation and anchoring convention.

Lemma 5.2 (Rational integration without a new letter)**.** *Let F be a finite sum of rational multiples of words ending in 1 and of the empty word. Suppose every rational factor has poles only at $0, 1$ and all word lengths are at most w. If F is holomorphic at zero, its primitive vanishing at zero is a rational linear combination of words ending in 1 of lengths at most $w+1$ and the empty word. It can be constructed with exact rational operations and finite differentiation. The same holds for $F(z)/z$ if $F(0) = 0$.*

Proof. For a rational function with poles only at $0, 1$, ordinary partial fractions give the exact Hermite decomposition

$$R(z) = Q'(z) + \frac{\alpha}{z} + \frac{\beta}{1-z}, \qquad Q \in K(z), \quad \alpha, \beta \in K. \tag{5.7}$$

Polynomial terms and poles of order at least two integrate into Q. For a nonempty word $\omega = a\nu$, integration by parts gives

$$\int R I_\omega \, \mathrm{d}z = Q I_\omega + \alpha I_{0\omega} + \beta I_{1\omega} - \int Q f_a I_\nu \, \mathrm{d}z. \tag{5.8}$$

The remaining integral has a strictly shorter word and a rational factor with the same possible poles. Repeated application terminates. The two prepended-letter terms have length at most $w+1$; every other term has no greater length. For the empty word the primitive is $Q + \alpha \log z - \beta \log(1-z)$.

Combine equal words and simplify rational coefficients before taking an endpoint. The constructed primitive is a meromorphic germ plus a constant multiple of $\log z$, since all other retained words are analytic at zero. Its derivative is the assumed holomorphic F. Therefore its Laurent principal part and its $\log z$ coefficient vanish. The remaining constant is determined by finitely many Taylor coefficients: each rational factor has a finite pole order. Subtract that constant. This gives the unique anchored analytic primitive and proves the claim. The case F/z follows because it is holomorphic under the additional hypothesis. This is the two-letter specialization of the standard hyperlogarithmic integration method [8]. □

For $\theta = z\, d/dz$ and an integer $h \geq 0$, define

$$\mathcal{J}_h F(z) = z^{-h} \int_0^z t^{h-1} F(t) \, \mathrm{d}t. \tag{5.9}$$

If $h = 0$, require $F(0) = 0$. If $h > 0$ and F is analytic, the apparent singularity is removable. The coefficient identity is

$$\mathcal{J}_h \left(\sum_{M \geq 1} a_M z^M \right) = \sum_{M \geq 1} \frac{a_M}{M+h} z^M. \tag{5.10}$$

Differentiation by θ does not increase word length; one application of $\mathcal{J}_h$ increases it by at most one. There is no unspecified integration constant in (5.9).

Theorem 5.3 (Uniform finite polylogarithm alphabet)**.** *For every rational weight* (2.1) *and every* $r \geq 1$,

$$\boxed{G_r(z; R) \in \mathscr{L}_{W_r(\mathbf{s})}, \qquad |z| < 1.} \tag{5.11}$$

An explicit terminating construction is as follows. Expand the confluent determinant, merge shifts, put its coefficients in the partial-fraction form (5.1)*, and quasi-shuffle each harmonic product. Replace it by* (5.5)*. Replace multiplication by* $p(M)$ *by* $p(\theta)$*, and division by* $(M+h)^e$ *by* $\mathcal{J}_h^e$*. Apply* (5.7)*–*(5.8) *to remove all unevaluated integrals. The output has rational coefficients in* $K(z)$ *and no letters other than* $0, 1$*.*

Proof. A harmonic product of weight b gives a finite sum of polylogarithms of weight b divided by $1-z$. The empty product has the rational generator already specified. Polynomial coefficient operators do not increase weight. Each inverse-cutoff factor raises it by at most its order, by Lemma 5.2. Lemma 5.1 therefore proves the stated weight bound term by term. All operations are finite: the determinant has finite size, partial fractions are finite, the quasi-shuffle has finitely

many words, and the integration recursion strictly decreases word length. Identities (5.5) and (5.10) prove equality of all Taylor coefficients. Both sides are analytic in $|z| < 1$, so the constructed expression equals the original generator throughout that disk, with removable values at zero taken collectively. □

Corollary 5.4 (Every quadratic denominator uses only a dilogarithm)**.** *Suppose the reduced denominator of R has degree at most two and all its roots are nonpositive integers; the polynomial part is arbitrary. For every $r \geq 1$ there are rational functions $A_r, B_r, C_r, E_r \in K(z)$ with poles only at $0, 1$ such that, with $L = -\log(1-z)$,*

$$G_r(z; R) = A_r(z) + B_r(z)L + C_r(z)L^2 + E_r(z)\,\mathrm{Li}_2(z). \tag{5.12}$$

Thus the adjacent-two-pole dilogarithm phenomenon extends to arbitrarily separated integer poles, a repeated double pole, and any polynomial part.

Proof. The possible nonempty pole profiles are (1), (1, 1), and (2). Their stabilized weights are at most two. The positive-index words of weight at most two are $\mathrm{Li}_1 = L$, Li_2, and $\mathrm{Li}_{1,1} = L^2/2$. Apply Theorem 5.3. □

Corollary 5.5 (Euler primitives and a finite differential module)**.** *For every integer $e \geq 0$,*

$$\sum_{M \geq 1} \frac{D_{r,M}(R)}{M^e} z^M = \mathcal{J}_0^e G_r(z; R) \in \mathscr{L}_{W_r+e}. \tag{5.13}$$

All generators G_r for a fixed weight R lie in a common differential module over $K(z)$ spanned by at most 2^{W_∞} functions. Consequently each individual G_r satisfies a nonzero scalar linear differential equation over $K(z)$ of order at most 2^{W_∞}.

Proof. The first statement follows by anchored iteration. There are 2^{w-1} compositions of positive weight w, so the empty word and all words of weight at most W_∞ give 2^{W_∞} generators. Differentiation maps their span into itself by the word derivative rule and the product rule for rational factors. The first $2^{W_\infty} + 1$ derivatives of an individual element are therefore linearly dependent over $K(z)$. This proves the scalar equation claim. It does *not* assert that the same scalar equation annihilates all ranks; rational prefactors vary with r. □

6 Several simple poles: fixed-size highest coefficients

For this section omit the polynomial part and write

$$R(x) = \sum_{j=1}^{d} \frac{c_j}{x + a_j} = \frac{P_*(x)}{L(x)}, \quad L(x) = \prod_{j=1}^{d} (x + a_j), \quad P_*(x) = \sum_{j=1}^{d} c_j \frac{L(x)}{x + a_j}, \tag{6.1}$$

where a_j are distinct nonnegative integers and $c_j \neq 0$. After (2.5), let X stand for H_M. The formal determinant is a polynomial in X with coefficients rational in M.

Theorem 6.1 (The top coefficient is a polynomial-weight determinant)**.** *For $r \geq d$, put $q = r - d$ and $T_{r,M} = [X^d] D_{r,M}(R)$. Then*

$$\boxed{T_{r,M} = \left(\prod_{j=1}^{d} c_j \right) \prod_{i<j} (a_j - a_i)^2 \det \left(\sum_{n=1}^{M} n^{i+j} L(n) P_*(n) \right)_{0 \leq i,j < q}.} \tag{6.2}$$

The determinant of size zero is one. In particular the highest harmonic coefficient has no rational cutoff poles.

Proof. The coefficient of X in the moment matrix is $\sum_j c_j v_j v_j^{\mathsf{T}}$, where $v_j = (1, -a_j, \dots, (-a_j)^{r-1})^{\mathsf{T}}$. Use the monic polynomial basis

$$1, x, \dots, x^{d-1}, L(x), xL(x), \dots, x^{q-1}L(x).$$

Its triangular coefficient matrix has determinant one. Each of the last q basis vectors vanishes at every $-a_j$, so the harmonic matrix has support only in the first d rows and columns. Its $d \times d$ determinant is $\prod_j c_j \prod_{i<j}(a_j - a_i)^2$. The coefficient of X^d must select this entire block. The complementary moment matrix sums $x^{i+j}L(x)^2R(x) = x^{i+j}L(x)P_*(x)$, a polynomial weight. This proves (6.2), including its normalization. □

For completeness, the monic uniform Hahn polynomial on $1, \dots, M$ can be written without hypergeometric denominators as

$$p_j(x; M) = \binom{2j}{j}^{-1} \sum_{\ell=0}^{j} (-1)^{j+\ell} \binom{j}{\ell} \binom{j+\ell}{\ell} (M-1-\ell)^{\underline{j-\ell}} (x-1)^{\underline{\ell}}. \tag{6.3}$$

For $j < M$ it is orthogonal with unit weights and has squared norm

$$h_j(M) = \frac{\Gamma(M+j+1)}{(2j+1)\Gamma(M-j)\binom{2j}{j}^2}. \tag{6.4}$$

These are the uniform specialization of the classical finite Hahn formulas [10]. They imply $\det(\sum_{n=1}^{M} n^{i+j})_{i,j<q} = \prod_{j<q} h_j(M)$ by a monic triangular change of basis. The endpoint identity used in Theorem 4.1 follows by putting $x = 0$ in (6.3) and applying the finite binomial Vandermonde identity.

Corollary 6.2 (At most $2d-1$ Hahn rows, at every particle count)**.** *Let $W(x) = L(x)P_*(x) = \lambda \prod_{i=1}^{e}(x - b_i)$ have its actual nonzero leading coefficient λ and degree $e \le 2d-1$. For distinct roots and $M \ge q + e$,*

$$\begin{aligned} T_{r,M} = \left(\prod_j c_j\right) \prod_{i<j} (a_j - a_i)^2 \lambda^q (-1)^{eq} \mathcal{U}_q(M) \\ \times \frac{\det(p_{q+j-1}(b_i; M))_{1 \le i,j \le e}}{\prod_{i<j}(b_j - b_i)}. \end{aligned} \tag{6.5}$$

Repeated roots are handled by the confluent divided-difference limit. The compressed formula (6.2) remains the definition at every smaller cutoff and whenever the leading coefficient changes degree. Thus the highest harmonic coefficient has a determinant of fixed size at most $2d-1$, independent of r.

Proof. For a measure ν and its monic orthogonal polynomials, the Christoffel determinant identity is

$$D_q\left[\prod_{i=1}^{e}(x - b_i)\nu\right] = (-1)^{eq} D_q[\nu] \frac{\det(p_{q+j-1}(b_i))_{i,j=1}^{e}}{\prod_{i<j}(b_j - b_i)}.$$

One proof appends the e fixed nodes b_i to the Vandermonde in the q integration nodes. The cross factors are $\prod_{i,j}(b_i - x_j)$, giving the stated sign. Replace monomials by monic orthogonal polynomials and integrate the first q rows against the companion Vandermonde. Orthogonality leaves the product of the first q squared norms and the displayed e rows. All steps are finite for the discrete measure here. This is the standard Christoffel transformation [7]. Applying it to (6.2), and extracting the constant factor λ^q, proves the result. Coalescing roots replaces evaluation rows by derivative rows divided by the corresponding factorials, because the Vandermonde quotient is a symmetric polynomial in the roots. □

6.1 Exceptional residues as an exact matrix spectrum

Let $M \geq q$ and $q > 0$. For real $a_j \geq 0$ define positive definite $q \times q$ matrices

$$B_j = \left(\sum_{n=1}^{M} n^{u+v} \frac{L(n)^2}{n + a_j} \right)_{0 \leq u,v < q}. \tag{6.6}$$

For a nonzero polynomial f of degree below q, its quadratic form is a positive sum of $f(n)^2$ over at least q distinct nodes. Hence B_j is positive definite.

Theorem 6.3 (Residue hypersurface and real one-parameter slices)**.** *Apart from the explicit factors $\prod_j c_j$ in (6.2), the highest-coefficient exceptional locus is exactly*

$$\det \left(\sum_{j=1}^{d} c_j B_j \right) = 0. \tag{6.7}$$

The polynomial on the left is nonzero when every $\operatorname{Im} c_j > 0$, and also when every $c_j > 0$ is real. Fix positive real values for all residues except $c_i = \tau$, and set $A_i = \sum_{j \neq i} c_j B_j$. If $d \geq 2$, then all exceptional values of τ in (6.7) are

$$\tau = -\lambda_1, \ldots, -\lambda_q, \tag{6.8}$$

where $\lambda_j > 0$ are the eigenvalues of $B_i^{-1/2} A_i B_i^{-1/2}$, counted with multiplicity. No simplicity assertion is made without additional hypotheses.

Proof. The compressed determinant in (6.2) is exactly $\det \sum_j c_j B_j$. If the imaginary parts are positive and a nonzero complex vector v were in its kernel, taking the imaginary part of $v^*(\sum_j c_j B_j)v$ would give $\sum_j \operatorname{Im} c_j \, v^* B_j v = 0$, a contradiction. The positive real case is immediate. This is the standard positive-matrix proof of determinantal stability [9], specialized to the present compressed measure. Finally

$$\det(A_i + \tau B_i) = \det B_i \prod_{j=1}^{q} (\tau + \lambda_j),$$

and A_i is positive definite for $d \geq 2$. This proves the exact spectral description. The extra zero $\tau = 0$ of the full top coefficient changes the active pole set and is kept separate. □

7 Exact endpoint cancellation equations

The highest coefficient in (6.2) is polynomial in M, but lower harmonic coefficients can have cutoff poles. They must be retained before forming a generating function. For the simple-pole weight (6.1), polynomial division gives the entirely explicit moments

$$\begin{aligned} \mu_j(M, X) = \sum_{a \in A} c_a \Bigg\{ & \sum_{v=0}^{j-1} (-a)^{j-1-v} S_v(M) \\ & + (-a)^j \left(X + \sum_{b=1}^{a} \frac{1}{M+b} - H_a \right) \Bigg\}. \end{aligned} \tag{7.1}$$

The inner polynomial sum is empty if $j = 0$.

For $1 \leq h \leq \max A$, put

$$A_h^+ = \{a \in A : a \geq h\}, \qquad d_h = |A_h^+|, \qquad v_a = (1, -a, \ldots, (-a)^{r-1})^{\mathsf{T}}.$$

Let $\boldsymbol{\mu}$ denote the formal moment matrix from (7.1), and define its regular part at $M = -h$ by

$$B_h(M, X) = \boldsymbol{\mu}(M, X) - \frac{1}{M+h} \sum_{a \in A_h^+} c_a v_a v_a^{\mathsf{T}}. \tag{7.2}$$

Every entry is analytic in M near $-h$ after its removable cancellations are simplified.

Theorem 7.1 (Every principal coefficient from a finite bordered jet)**.** *The order of the formal cutoff pole at $M = -h$ is at most $\min(r, d_h)$. Write its principal part as $\sum_{\ell=1}^{\min(r,d_h)} P_{h,\ell}(X)/(M+h)^\ell$. For a subset $S \subset A_h^+$ of size k, let V_S have columns v_a for $a \in S$. Then*

$$\boxed{P_{h,\ell}(X) = \sum_{k=\ell}^{\min(r,d_h)} \sum_{\substack{S \subset A_h^+ \\ |S|=k}} \frac{(-1)^k \prod_{a \in S} c_a}{(k-\ell)!} \, \partial_M^{k-\ell} \det \begin{pmatrix} B_h(M, X) & V_S \\ V_S^{\mathsf{T}} & 0 \end{pmatrix}\Bigg|_{M=-h}}. \tag{7.3}$$

Thus the pole cancels identically as a formal harmonic expression if and only if all of these polynomials in X vanish. More generally, its exact order is the largest ℓ with $P_{h,\ell} \neq 0$. Moreover

$$\deg_X P_{h,\ell} + \ell \leq \min(r, d). \tag{7.4}$$

Proof. For any square matrix B and vectors v_a, the rank-one update expansion is

$$\det\left(B + \sum_a t_a v_a v_a^{\mathsf{T}}\right) = \sum_S (-1)^{|S|} \left(\prod_{a \in S} t_a\right) \det \begin{pmatrix} B & V_S \\ V_S^{\mathsf{T}} & 0 \end{pmatrix}. \tag{7.5}$$

For the empty subset the bordered determinant is $\det B$. One can prove the identity first when B is invertible by the Schur complement and the principal-minor expansion of $\det(I + \operatorname{diag}(t) V^{\mathsf{T}} B^{-1} V)$. Both sides are polynomials in the entries of B, so the identity holds without invertibility. Terms with $|S| > r$ vanish.

Apply (7.5) with $B = B_h$ and $t_a = c_a/(M+h)$. Taylor expansion of each analytic bordered determinant gives (7.3). It also proves the maximal pole order and the necessary-and-sufficient cancellation criterion. Finally, in the simple-pole case the determinant is multiaffine in the original $h_{a,1}$ and contains at most $\min(r, d)$ such factors. Selecting ℓ inverse-cutoff factors leaves at most $\min(r, d) - \ell$ harmonic factors. Equivalently use Lemma 5.1 with all $s_a = 1$. This proves (7.4). □

This is an exact finite classification by explicit polynomial equations, not an intrinsic formula for the first particle count at which every endpoint disappears. Such a threshold, expressed only in pole locations and residues without constructing the jets, remains a further question. The theorem nevertheless gives every endpoint contribution required by the compiler, including cases in which a naive residue-rank estimate is strict.

7.1 A complete three-pole example

Take

$$R(n) = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2}, \qquad r = 3. \tag{7.6}$$

The exact formal determinant has the following compact polynomial-plus-principal-part representation:

$$\begin{aligned} D_{3,M}(R) = {} & 4H_M^3 + A_2(M) H_M^2 + A_1(M) H_M + A_0(M) \\ & + \frac{8H_M^2 - 36H_M - 19/2}{M+1} + \frac{4H_M - 18}{(M+1)^2} + \frac{4H_M^2 - 26H_M + 13}{M+2}, \end{aligned} \tag{7.7}$$

where

$$\begin{aligned}
A_2(M) &= \tfrac{9}{2}M^4 + 27M^3 + \tfrac{111}{2}M^2 + 45M - 10,\\
A_1(M) &= \tfrac{3}{8}M^6 + \tfrac{27}{8}M^5 - \tfrac{45}{4}M^4 - \tfrac{753}{8}M^3 - \tfrac{1199}{8}M^2 - \tfrac{69}{2}M + 45,\\
A_0(M) &= -\tfrac{23}{16}M^6 - \tfrac{129}{16}M^5 + \tfrac{219}{16}M^4 + \tfrac{1527}{16}M^3 + \tfrac{495}{8}M^2 - \tfrac{169}{4}M + 21.
\end{aligned} \tag{7.8}$$

These formulas follow by inserting (7.6) in (7.1) and expanding a 3×3 determinant; the accompanying exact checks also compare them to the original finite sums. The top coefficient is 4, precisely $\prod_{i<j}(a_j - a_i)^2$ at $a_j = 0, 1, 2$. The double cutoff pole at $M = -1$ shows why retaining only simple endpoint poles is incorrect for three simple summand poles. This is a counterexample to that possible extrapolation, not a claim that the adjacent-pole source made it.

Let $F_j(z) = \sum_{M\ge1} H_M^j z^M$, with $F_0 = z/(1-z)$. The finite product rules give

$$\begin{aligned}
F_1 &= \frac{\mathrm{Li}_1(z)}{1-z},\\
F_2 &= \frac{\mathrm{Li}_2(z) + 2\,\mathrm{Li}_{1,1}(z)}{1-z},\\
F_3 &= \frac{\mathrm{Li}_3(z) + 3\,\mathrm{Li}_{2,1}(z) + 3\,\mathrm{Li}_{1,2}(z) + 6\,\mathrm{Li}_{1,1,1}(z)}{1-z}.
\end{aligned} \tag{7.9}$$

An explicit identity for the three-pole generator is therefore

$$\begin{aligned}
G_3(z;R) = {} & 4F_3 + A_2(\theta)F_2 + A_1(\theta)F_1 + A_0(\theta)F_0\\
& + \mathcal{J}_1(8F_2 - 36F_1 - \tfrac{19}{2}F_0) + \mathcal{J}_1^2(4F_1 - 18F_0)\\
& + \mathcal{J}_2(4F_2 - 26F_1 + 13F_0).
\end{aligned} \tag{7.10}$$

Here each $\mathcal{J}_h$ is the explicitly anchored integral (5.9). The supplied compiler removes them completely and returns eight rationally weighted words of length at most three. The fully expanded coefficients are recorded in `data/compiled_examples.json`; (7.10) is the more readable exact expression. The derivative operators are of finite order and act on the entire functions immediately following them.

8 A small exact identity with a rank-independent extension

8.1 The triple pole and its weight-four primitive

For the triple pole at the origin and two particles, put

$$K_M = H_M^{(1)} H_M^{(3)} - \big(H_M^{(2)}\big)^2. \tag{8.1}$$

The diagonal terms in the two quasi-shuffles cancel. Consequently

$$\boxed{\sum_{M\ge1} K_M z^M = \frac{\mathrm{Li}_{3,1}(z) + \mathrm{Li}_{1,3}(z) - 2\,\mathrm{Li}_{2,2}(z)}{1-z}.} \tag{8.2}$$

This concrete weight-four identity is also the first nontrivial member of the spectral family in the next section. An anchored primitive is

$$\int_0^z \frac{1-t}{t} \sum_{M\ge1} K_M t^M \,\mathrm{d}t = \mathrm{Li}_{4,1}(z) + \mathrm{Li}_{2,3}(z) - 2\,\mathrm{Li}_{3,2}(z). \tag{8.3}$$

The right side is single-valued on the disk under the power-series branch convention. There is no additive constant because both sides vanish at zero.

Proposition 8.1 (An ordinary convergent zeta evaluation)**.**

$$\boxed{\sum_{M=1}^{\infty} \frac{H_M^{(1)} H_M^{(3)} - (H_M^{(2)})^2}{M(M+1)} = \frac{35}{2}\zeta(5) - 9\zeta(2)\zeta(3).} \tag{8.4}$$

Proof. The numerator is $O(1 + \log M)$, so the sum is absolutely convergent. Alternatively (2.3) expresses it as $\sum_{n>m\geq 1}(n-m)^2/(n^3m^3)$ up to the cutoff; all terms are nonnegative. At $z = 1$, integrating (8.2) as in (8.3) therefore gives

$$\zeta(4,1) + \zeta(2,3) - 2\zeta(3,2). \tag{8.5}$$

Here $\zeta(a,b) = \sum_{n>m\geq 1} n^{-a}m^{-b}$.

For transparency, the classical weight-five reduction can be checked with three identities. Write $P = \zeta(2)\zeta(3)$ and $Z = \zeta(5)$. The height-one Gamma identity gives $\zeta(4,1) = 2Z - P$. One way to see its exact coefficient is to use

$$1 - \frac{\Gamma(1-x)\Gamma(1-y)}{\Gamma(1-x-y)} = \sum_{m,n\geq 1} \zeta(m+1, \{1\}^{n-1})x^m y^n. \tag{8.6}$$

For a direct normalization check, sum the height-one iterated-integral formula over m, n to obtain

$$y\int_0^1 (t^{-x} - 1)(1-t)^{-y-1}\,\mathrm{d}t = y\{B(1-x,-y) - B(1,-y)\} = 1 - \frac{\Gamma(1-x)\Gamma(1-y)}{\Gamma(1-x-y)}.$$

The difference integral is analytic for small x, y; the separated beta terms may first be evaluated with $\operatorname{Re} y < 0$ and then continued. This is consistent with the height-one Gamma convention in the repository [5]. Its logarithmic Gamma expansion gives coefficient x^3y^2 equal to $2\zeta(5) - \zeta(2)\zeta(3)$: the linear exponent contributes $2\zeta(5)$ and the product of its weight-two and weight-three terms contributes $-\zeta(2)\zeta(3)$.

The finite-sum stuffle, passed to the absolutely convergent limit, gives $P = \zeta(2,3) + \zeta(3,2) + Z$. The iterated-integral shuffle of words 01 and 001 gives $P = \zeta(2,3) + 3\zeta(3,2) + 6\zeta(4,1)$; the multiplicities follow by listing the ten order-preserving interleavings. Solving yields

$$\zeta(3,2) = 3P - \tfrac{11}{2}Z, \qquad \zeta(2,3) = \tfrac{9}{2}Z - 2P.$$

Substitute in (8.5). This proves (8.4) by exact identities, without a numerical-relation search. □

The reduction of (8.5) is classical Euler-sum algebra. The structural extension here is that every determinant $D_{r,M}(n^{-3})$, for arbitrary r, admits a generator in the same weight-four module, with the explicit nonzero coefficient (4.3). For example the next determinant is

$$\begin{aligned} D_{3,M}(n^{-3}) = {} & S_1(M)\{H_M^{(1)}H_M^{(3)} - (H_M^{(2)})^2\} \\ & - M^2 H_M^{(3)} + 2M H_M^{(1)} H_M^{(2)} - (H_M^{(1)})^3. \end{aligned} \tag{8.7}$$

Its generator follows by quasi-shuffling just the displayed products and replacing M by θ. Higher rank changes the rational coefficients and differential operators, not the maximal word length.

8.2 A shifted double pole with an ordinary zeta endpoint

The repeated-pole case of Corollary 5.4 gives another small identity. Set

$$B_M = M\{H^{(2)}_{M+1} - 1\} - (H_{M+1} - 1)^2, \qquad L = -\log(1-z).$$

A monic change from the basis $1, n$ to $1, n+1$ shows that B_M is $D_{2,M}((n+1)^{-2})$.

Proposition 8.2 (Shifted double-pole generator and Euler primitive)**.** *For $|z| < 1$,*

$$\boxed{\sum_{M\geq 1} B_M z^M = \frac{(3z-2)\operatorname{Li}_2(z) + (1-z)(3L - L^2) - z}{z(1-z)^2}.} \tag{8.8}$$

The value at zero is removable. Moreover

$$\boxed{\sum_{M\geq 1} \frac{M\{H^{(2)}_{M+1} - 1\} - (H_{M+1} - 1)^2}{M(M+1)(M+2)} = \frac{5}{2}\zeta(3) - \frac{3}{2}\zeta(2) - \frac{1}{2}.} \tag{8.9}$$

Proof. Let $U = \sum_{M\geq 1}(H^{(2)}_{M+1} - 1)z^M$ and $V = \sum_{M\geq 1}(H_{M+1} - 1)^2 z^M$. The elementary shift of (5.5) and the quadratic quasi-shuffle give

$$U = \frac{\operatorname{Li}_2(z)}{z(1-z)} - \frac{1}{1-z}, \qquad V = \frac{\operatorname{Li}_2(z) + L^2 - 2L}{z(1-z)} + \frac{1}{1-z}.$$

Their combination $\theta U - V$ is precisely (8.8).

The determinant is a nonnegative finite Vandermonde sum and is $O(M)$. Its endpoint series is therefore absolutely convergent. Since

$$\frac{1}{2}\int_0^1 z^{M-1}(1-z)^2\,\mathrm{d}z = \frac{1}{M(M+1)(M+2)},$$

it equals $\frac{1}{2}\int_0^1 N(z)/z^2\,\mathrm{d}z$, where $N(z) = (3z-2)\operatorname{Li}_2(z) + (1-z)(3L - L^2) - z$. Here $N(z)/z \to 0$ at zero and $N(1) = \zeta(2) - 1$. Direct differentiation gives

$$N'(z) = 3\operatorname{Li}_2(z) + L^2 - 2L + 2 - \frac{2L}{z}.$$

Integrate by parts once. Use $\int_0^1 \operatorname{Li}_2(z)\,\mathrm{d}z/z = \zeta(3)$, $\int_0^1 L\,\mathrm{d}z/z = \zeta(2)$, and $\int_0^1 L^2\,\mathrm{d}z/z = 2\zeta(2,1) = 2\zeta(3)$. The last equality follows also from coefficient xy^2 in (8.6). The remaining terms must be kept together:

$$\int_0^1 \left(\frac{1}{z} - \frac{L}{z^2}\right)\mathrm{d}z = \left[\frac{L}{z} - L\right]_0^1 = -1.$$

Thus the integral is $\frac{1}{2}\{-\zeta(2) + 1 + 5\zeta(3) - 2\zeta(2) - 2\}$, which proves (8.9). □

9 A spectral master identity at every rank

The integer-pole theorem has a useful analytic companion that should not be confused with differentiating its formal weight bound. Define, for arbitrary complex s,

$$\mathcal{K}_{r,M}(s) = \det\bigl(H_M^{(s-i-j)}\bigr)_{0\leq i,j<r}. \tag{9.1}$$

For a finite cutoff, each entry is an entire function of s, with $n^{-s} = \exp(-s \log n)$ using the real logarithm. In this section $\mathrm{Li}_{a_1,\dots,a_r}(z)$ is also allowed to have arbitrary complex indices; its defining nested series is still entire in all indices when $|z| < 1$.

Let $\mathfrak{S}_r$ act on $\{0, \dots, r-1\}$. For $\sigma, \tau \in \mathfrak{S}_r$, put

$$a_j(s; \sigma, \tau) = s - \sigma(j) - \tau(j), \qquad 0 \le j < r. \tag{9.2}$$

Theorem 9.1 (All-rank spectral polylogarithm and zeta identities)**.** *For every $s \in \mathbb{C}$ and $|z| < 1$,*

$$\boxed{\sum_{M \ge 1} \mathcal{K}_{r,M}(s) z^M = \frac{1}{1-z} \sum_{\sigma,\tau \in \mathfrak{S}_r} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \, \mathrm{Li}_{a_0(s),\dots,a_{r-1}(s)}(z).} \tag{9.3}$$

Both sides are entire in s and can be differentiated to every order on compact subsets. If $\operatorname{Re} s > 2r-2$, then the ordinary endpoint sum is absolutely convergent and equals

$$\boxed{\sum_{M \ge 1} \frac{\mathcal{K}_{r,M}(s)}{M(M+1)} = \sum_{\sigma,\tau \in \mathfrak{S}_r} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \zeta(a_0(s)+1, a_1(s), \dots, a_{r-1}(s)).} \tag{9.4}$$

The endpoint identity is holomorphic in this half-plane and also admits all fixed-order spectral derivatives there.

Proof. Apply (2.3) with $R(n) = n^{-s}$. Reverse the node order to $n_0 > \cdots > n_{r-1}$, which does not change the square of the Vandermonde. Expand that square as two determinants:

$$\Delta(n_0, \dots, n_{r-1})^2 = \sum_{\sigma,\tau} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \prod_{j=0}^{r-1} n_j^{\sigma(j)+\tau(j)}.$$

Sum the cutoff $M \ge n_0$ geometrically. This gives (9.3), including the absence of an $r!$ factor. For s on a compact set, all finite harmonic moments and their fixed spectral derivatives have polynomial growth in M times powers of $\log(2+M)$. The factor z^M supplies a locally uniform summable majorant, proving entireness and every fixed-order differentiation. The same argument applies to each nested polylogarithm.

For the endpoint, the cutoff tail is exactly

$$\sum_{M=n_0}^{\infty} \frac{1}{M(M+1)} = \frac{1}{n_0}.$$

To justify all absolute rearrangements in the claimed half-plane, use the elementary rearrangement inequality on the decreasing nodes: for every permutation σ, $\prod_j n_j^{\sigma(j)} \le \prod_j n_j^{r-1-j}$. It follows by successively swapping any inverted pair of exponents. After both Vandermonde expansions and the extra factor $1/n_0$, every absolute summand is dominated by

$$n_0^{-\operatorname{Re} s + 2r - 3} \prod_{j=1}^{r-1} n_j^{-\operatorname{Re} s + 2r - 2 - 2j}.$$

When $\operatorname{Re} s > 2r-2$, all the positive decay exponents in this expression exceed one. Dropping the ordering gives a product of convergent ordinary zeta series. A compact sub-half-plane permits additional fixed powers of logarithms by slightly decreasing those exponents. This proves convergence, (9.4), and all stated derivatives. □

At a positive integer s, (9.3) can contain nonpositive indices if r is large. Theorem 5.3 removes them constructively and yields the uniform alphabet of maximal weight $\lfloor (s+1)^2/4 \rfloor$, independent of r. The unreduced spectral formula is not itself a minimal-weight representation. At a general noninteger s, or after order differentiation, the finite integer-weight alphabet theorem is not being asserted.

10 Generalized Stieltjes constants and exact spectral jets

Use the Laurent convention

$$\zeta(1+u,a) = \frac{1}{u} + \sum_{q\ge 0} \frac{(-1)^q}{q!}\gamma_q(a)u^q. \tag{10.1}$$

For $p \in \mathbb{C}$, define the finite logarithmic harmonic sums

$$L_{q,p}(M) = \sum_{n=1}^{M} \frac{\log^q n}{n^p} = (-1)^q \partial_p^q H_M^{(p)}. \tag{10.2}$$

The Hurwitz difference $H_M^{(p)} = \zeta(p) - \zeta(p, M+1)$ is entire in p because its two residues at $p=1$ cancel. Comparing (10.1) gives the ordinary finite identity

$$J_q(M) := \gamma_q(1) - \gamma_q(M+1) = L_{q,1}(M). \tag{10.3}$$

This is a difference of Stieltjes constants, not a value assigned to one of the separately singular terms $\zeta^{(q)}(1,a)$. For $p \ge 2$ integral,

$$L_{q,p}(M) = (-1)^q\{\zeta^{(q)}(p) - \zeta^{(q)}(p, M+1)\}. \tag{10.4}$$

At $q=0$ the same harmonic variables are polygamma differences:

$$H_M^{(p)} = \frac{(-1)^{p-1}}{(p-1)!}\{\psi^{(p-1)}(M+1) - \psi^{(p-1)}(1)\}, \qquad p \ge 1. \tag{10.5}$$

For $p=1$ this uses $\psi^{(0)} = \psi$ and is the usual digamma shift. These standard Hurwitz and polygamma identities are consistent with DLMF conventions [11]; the finite sums above also prove the needed cancellations directly.

10.1 A complete rank-two Stieltjes family

Define a holomorphic function for $\operatorname{Re} s > 2$ by

$$\mathcal{Z}(s) = \zeta(s+1, s-2) + \zeta(s-1, s) - 2\zeta(s, s-1). \tag{10.6}$$

The double-zeta series in this formula all converge absolutely on that half-plane.

Theorem 10.1 (All-order Stieltjes–harmonic endpoint identity)**.** *For every integer $q \ge 0$,*

$$\boxed{\sum_{M=1}^{\infty} \frac{1}{M(M+1)} \sum_{j=0}^{q} \binom{q}{j} \{L_{j,3}(M)J_{q-j}(M) - L_{j,2}(M)L_{q-j,2}(M)\} = (-1)^q \mathcal{Z}^{(q)}(3).} \tag{10.7}$$

Every series here is an ordinary absolutely convergent series. At $q = 0$ it is (8.4). For $|z| < 1$ the corresponding cutoff generator is

$$\begin{aligned}\sum_{M\geq 1} z^M \sum_{j=0}^{q} \binom{q}{j} \{L_{j,3}(M)J_{q-j}(M) - L_{j,2}(M)L_{q-j,2}(M)\} \\ = \frac{(-1)^q}{1-z}\,\partial_s^q\{\mathrm{Li}_{s,s-2}(z) + \mathrm{Li}_{s-2,s}(z) - 2\,\mathrm{Li}_{s-1,s-1}(z)\}|_{s=3}\,.\end{aligned} \tag{10.8}$$

Proof. For $r = 2$, (9.1) is $H_M^{(s)}H_M^{(s-2)} - (H_M^{(s-1)})^2$. Differentiate this finite expression q times at $s = 3$, multiply by $(-1)^q$, and use Leibniz's rule, (10.2), and (10.3). The result is precisely the inner binomial sum. Apply the corresponding derivative to (9.4) and (9.3). Theorem 9.1 justifies both interchanges. Directly, $L_{j,2}$ and $L_{j,3}$ remain bounded in M for each fixed j, whereas $J_j(M) = O(1 + \log^{j+1}(2 + M))$. This also checks absolute convergence of the displayed endpoint series. □

For example, the first logarithmic identity is

$$\sum_{M\geq 1} \frac{L_{1,3}(M)H_M^{(1)} + H_M^{(3)}\{\gamma_1(1) - \gamma_1(M+1)\} - 2H_M^{(2)}L_{1,2}(M)}{M(M+1)} = -\mathcal{Z}'(3). \tag{10.9}$$

This is an exact relation between generalized Stieltjes constants, Hurwitz-zeta derivatives, and depth-two zeta derivatives. It is not obtained by differentiating the isolated evaluation $\mathcal{Z}(3) = \frac{35}{2}\zeta(5) - 9\zeta(2)\zeta(3)$. That evaluation holds at one value of s; it does not determine $\mathcal{Z}'(3)$ or higher derivatives. Promoting a special-value identity to a parameter identity without proof would give an invalid reduction of (10.9).

10.2 The all-rank version at the Stieltjes boundary

Let $s_0 = 2r - 1$ and $q \geq 0$. All integer indices $s_0 - i - \sigma(i)$ in the following formula are positive. Expanding the determinant and its derivative gives

$$\mathcal{T}_{r,q}(M) = \sum_{\sigma\in\mathfrak{S}_r} \mathrm{sgn}(\sigma) \sum_{q_0+\cdots+q_{r-1}=q} \frac{q!}{q_0!\cdots q_{r-1}!} \prod_{i=0}^{r-1} L_{q_i,s_0-i-\sigma(i)}(M). \tag{10.10}$$

With $a_j(s) = a_j(s;\sigma,\tau)$ as in (9.2), Theorem 9.1 proves

$$\begin{aligned}\sum_{M\geq 1} \frac{\mathcal{T}_{r,q}(M)}{M(M+1)} = (-1)^q\,\partial_s^q \Bigg\{ \sum_{\sigma,\tau\in\mathfrak{S}_r} \mathrm{sgn}(\sigma)\,\mathrm{sgn}(\tau) \\ \times\zeta(a_0(s)+1, a_1(s), \ldots, a_{r-1}(s))\Bigg\}\Big|_{s=s_0}\,.\end{aligned} \tag{10.11}$$

Formula (10.3) replaces every index-one factor by an actual Stieltjes difference. The other factors are (10.4); hence this is a finite all-rank, all-order specification of ordinary Stieltjes–zeta identities. There is at most one index-one factor in any product: it requires $i = \sigma(i) = r - 1$. At $q = 0$, the zeta terms on the right have total index weight $r^2 + 1$. Their possible further reduction to a chosen multiple-zeta basis is a distinct arithmetic problem, not needed for the identity.

10.3 Anchored spatial primitives and the Gamma endpoint

There is also a useful exact spatial primitive, stated here to fix its normalization rather than as a new special-function fact. For $a > -1$, let $J_q(M, a) = \gamma_q(a+1) - \gamma_q(M+a+1)$. For $-1 < \alpha < \beta$,

$$\begin{aligned}\int_\alpha^\beta J_q(M,a)\,\mathrm{d}a &= \frac{1}{q+1}\sum_{n=1}^{M}\{\log^{q+1}(n+\beta) - \log^{q+1}(n+\alpha)\} \\ &= \frac{(-1)^{q+1}}{q+1}\left[\zeta^{(q+1)}(0,a+1) - \zeta^{(q+1)}(0,M+a+1)\right]_{a=\alpha}^{a=\beta}.\end{aligned} \tag{10.12}$$

The proof integrates each finite logarithmic summand and differentiates the entire finite Hurwitz difference at zero. At $q = 0$ it reduces to

$$\int_\alpha^\beta \{\psi(M+a+1) - \psi(a+1)\}\,\mathrm{d}a = \left[\log\frac{\Gamma(M+a+1)}{\Gamma(a+1)}\right]_\alpha^\beta. \tag{10.13}$$

Positive real arguments fix the logarithm branch. These formulas may also be continued to a simply connected complex domain avoiding all $-1, -2, \ldots, -M$, with one consistent logarithm branch.

11 Verification, claim status, and source corrections

11.1 Exact checks and numerical diagnostics have different roles

The all-rank proofs above do not depend on fitting coefficient tables. The included programs independently check finite instances and the translation from formal identities to code. Their output is retained in machine-readable JSON and ordinary run logs.

Executed exact check	Count
Direct determinant versus compiled Taylor coefficient comparisons	99
Single-pole weighted-degree checks, orders 1–7, ranks 1–4	28
Extremal Gamma-product coefficient checks on the same grid	28
Several-simple-pole top-coefficient compression checks	8
Local confluent matrix factorizations using independent polynomial division	12
Endpoint bordered-jet principal coefficients	7
All-rank double-permutation spectral normalizations	24
Deliberately corrupted identities rejected	2

The compiler checks include separated simple poles, three simple poles, a shifted double pole, a mixture of double and simple poles, and a polynomial part. For each of nine examples, the constant coefficient and coefficients through z^{10} are compared exactly with the original finite-sum determinant. The maximum output word length is checked against W_r. The local factorization tests use polynomial long division of the rational summand rather than the compiler's shifted-binomial formula. The endpoint tests compare the bordered-jet formula to a separately expanded Laurent series.

The numerical suite uses 60 decimal digits and contains 28 diagnostics: original cutoff determinants against the compiled expressions; nonintegral and complex spectral parameters; Stieltjes differences through order three; four differentiated spectral generators; and independent Mellin

integrals for the weight-five double-zeta reduction. The suite uses a stated scaled tolerance of 10^{-48}. Recorded residuals, working precision, and truncation settings are in `data/numeric_checks.json`. These are floating-point diagnostics, not interval enclosures. A small residual alone is not used to prove any theorem, and no proof-assistant formalization is claimed.

11.2 What the package implements

`verification/compiler.py` implements the rational moment formula, finite quasi-shuffles, the two-letter integration recursion, and the anchored inverse operators. Its output is a dictionary of binary words and rational coefficients; an empty word means the rational term. The example file records this complete representation, not just numerical values. The finite algorithm works at arbitrary specified rank, but no polynomial-time complexity claim is made: symbolic expression size and partial-fraction expansion can grow substantially.

A minimal call is

```
from verification.compiler import compile_hankel, encode
result = compile_hankel(2, [(0, 3, 1)])
print(encode(result))
```

Each triple is `(nonnegative integer shift, pole order, residue)`. The output here is exactly (8.2) in binary-word notation. The program is intended for exact rational or symbolic coefficients; passing binary floating-point residues changes the nature of its arithmetic and is not part of the exact certificate claims.

11.3 Corrections and integration decisions

No false mathematical theorem was found in the particular inherited proof dependencies used here. The following are status corrections and precision safeguards, not accusations of errors in those theorems.

The gap-six report's Question Q4 still asks for the Gamma normalization constants K_m. The newer incoming Gamma-normalization package gives their explicit evaluation [4]; Q4 should point to that result instead of remaining an unspecialized open question. Similarly, the older mixed Gamma normal-form coefficient question is already settled by the universal Gamma archive [3] and should not be reopened on the strength of an older manuscript snapshot.

For the several-pole program, Questions Q7–Q9 can now reference the following specific advances: a fixed-size highest-coefficient formula for any finite simple pole set; exact finite endpoint cancellation jets; a uniform finite polylogarithm compiler including repeated integer poles; and the spectral Euler primitives proved here. A full intrinsic classification of all cutoff cancellations, or a necessary-and-sufficient criterion for *minimal classical* polylogarithm depth, remains open. Those broader questions should not be marked completely solved merely because a finite compiler exists.

Three conventions must survive integration. First, inverse-cutoff operators are anchored at zero, and principal parts are retained before summation. Second, all degree and weight statements concern formal harmonic coordinates; they do not assert independence of evaluated constants. Third, order derivatives of zeta functions are taken in actual parameter identities, not in an isolated special-value reduction. These distinctions are mathematically necessary.

12 Further research questions

Q1. Determine minimal functional weight and depth. Theorem 4.1 proves formal sharpness, not transcendental minimality. Which pole profiles force genuinely depth-two or higher functions after all functional identities and argument changes are allowed? In particular, classify the all-rank weights whose generators use only classical Li_j and logarithms.

Q2. Find a generic multiple-pole sharpness theorem. For a single pole the extremal coefficient is explicit. For several poles, characterize when the optimized confluent minors survive their compression and the merger of integer-shift harmonic variables. A coefficient formula indexed by the maximizing allocations k_a may distinguish generic sharpness from special residue cancellations.

Q3. Replace endpoint jets by intrinsic cutoff invariants. Formula (7.3) is a complete finite test, but it still uses matrices depending on r. Seek exact vanishing thresholds and factorizations directly from the integer dispersions of the poles, their residues, and the polynomial part. The three-pole double principal part in (7.7) is a first test case.

Q4. Study the exceptional residue hypersurface. Equation (6.7) describes it exactly and gives its positive-matrix slices. Determine its irreducible components, singular loci, and conditions for simple roots on a one-residue slice. Multiplicity cannot be discarded from the present theorem without a further argument.

Q5. Extend the compiler to rational and nonintegral shifts. For rational shifts, a roots-of-unity alphabet or a suitable cover is natural. Derive a precise analogue of W_r that tracks both the confluent multiplicities and the required alphabet. Arbitrary complex shifts should instead lead to a Lerch-type iterated-sum calculus.

Q6. Reduce the Stieltjes spectral derivatives arithmetically. The identities (10.7) and (10.11) leave explicit derivatives of multiple zeta functions. Seek genuine parameter identities reducing some of them to ordinary zeta or Hurwitz-zeta derivatives. Differentiating the weight-five evaluation at a single point is not an admissible shortcut.

Q7. Make the differential module small and explicit. The finite 2^{W_∞} generating set need not be minimal. Construct cyclic equations, differential relations, or a reduced unipotent system adapted to a fixed rational weight, and compare its rank across residue specializations. The target is an exact common module, not a falsely claimed single scalar equation for every particle count.

Q8. Understand confluence at the level of identities. Continuous merging of simple poles differentiates shift parameters and can introduce higher-weight harmonic variables. Explain precisely how the simple-pole identities pass to the repeated-pole capacities, including the passage from three simple poles to the weight-four triple-pole sector, while keeping the endpoint subtractions uniform.

Conclusion

A finite pole set supplies more than a finite harmonic rank. Its confluent derivative blocks determine an explicit weight allocation, and that allocation survives every rational cutoff correction needed for summation. This yields a finite, reproducible polylogarithm identity at every rank. The single-pole

coefficient (4.3) proves the formal invariant is attained and identifies its Gamma normalization. The fixed-size compression, endpoint jets, and spectral Stieltjes identities give complementary ways to extract exact consequences. They extend a concrete current repository question without confusing classical tools, newly proved combinations, finite evidence, and the larger arithmetic problems that remain.

References

[1] V. Reshetnikov, *ProveIt*, polylogarithm manuscript and incoming research packages. Repository observation at commit `505c62fddea6b4a9eb61fee0c76e6c021e9c7128`, 11 October 2026 local time. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt incoming research report, *Gap Six Harmonic Determinants*, archive `docs/incoming/ProveIt_Gap_Six_Harmonic_Determinants_2026-10-12.zip`. In particular `sections/02_complex_rank.tex`, `sections/03_hahn_and_polylogs.tex`, and `sections/05_audit_and_questions.tex`. Git blob `d7695b94c3117d7bdcb5e75fcf930c410e9c31b6` at the comparison commit.

[3] ProveIt incoming research report, *A Universal Beta Formula for Gamma Normal Forms*, archive `docs/incoming/ProveIt_Universal_Gamma_Normal_Forms_2026-10-11.zip`. README and result scope inspected. Git blob `b2a5e51a1b340683721bb803c20410fc3124d9de`.

[4] ProveIt incoming research report, *Gamma Determinants and Colored Stieltjes Moments*, archive `docs/incoming/ProveIt_Gamma_Normalizations_and_Colored_Moments_2026-10-12.zip`. Gamma determinant and normalization section inspected. Git blob `a7d997b5d80ab90623ca58864871eb3829089628`.

[5] ProveIt, *Stieltjes harmonic identities*, harmonic Gamma-ratio and height-one zeta development, source `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/stieltjes-harmonic-identities/sections/harmonic.tex`. Inspected Git blob `51108e8519d46af87dc3682956511fc525cecec3`.

[6] M. E. Hoffman, *Quasi-shuffle products*, Journal of Algebraic Combinatorics **11** (2000), 49–68. https://arxiv.org/abs/math/9907173.

[7] C. Krattenthaler, *Hankel determinants of linear combinations of moments of orthogonal polynomials, II*, The Ramanujan Journal **61** (2023), 597–627. https://doi.org/10.1007/s11139-021-00514-8.

[8] E. Panzer, *Algorithms for the symbolic integration of hyperlogarithms with applications to Feynman integrals*, Computer Physics Communications **188** (2015), 148–166. https://arxiv.org/abs/1403.3385.

[9] J. Borcea and P. Brändén, *The Lee–Yang and Pólya–Schur programs. II. Theory of stable polynomials and applications*. https://arxiv.org/abs/0809.3087. The positive-matrix stability argument needed here is proved in the text.

[10] NIST Digital Library of Mathematical Functions, Chapter 18, Sections 18.19–18.20, Hahn class and explicit representations. https://dlmf.nist.gov/18.20. Accessed 11 October 2026 local time.

[11] NIST Digital Library of Mathematical Functions, Section 25.11, Hurwitz zeta function, shifts, derivatives, and Gamma relations. https://dlmf.nist.gov/25.11. Accessed 11 October 2026 local time.